

Anjali Singh |

✉ anjali.singh@ischool.utexas.edu • 🌐 sites.google.com/view/singhanj13

Human-AI Interaction researcher and cognitive scientist with 8+ years experience developing and evaluating human-centered technologies across industry and academia. Experienced leading interdisciplinary teams in the design, development, large scale experimentation, and mixed-methods analysis of AI systems for information seeking, learning & tutoring.

Education

School of Information

Ph.D. in Information

University of Michigan, Ann Arbor

2019–2024

Mathematics and Computing

Integrated Bachelor and Master of Technology

Indian Institute of Technology, Delhi

2012–2017

Research and Work Experience

University of Texas at Austin | Postdoctoral Research Fellow

Enhancing Learning & Information Seeking with Generative AI

Sept'24–Current

- Designed and developed **interactive GenAI-powered search and note-taking tool**, that provides AI-generated metacognitive cues by proactively **monitoring user behavior to support trust, reasoning, and sensemaking** [Demo].
- Increased number of search queries** by 46% in GenAI-based search compared to baseline. **Increased user exploration breadth** by 88% and **improved sustained inquiry** behaviors by 60%, demonstrating significantly improved engagement.
- Defined and **operationalized evaluation metrics** such as persistent inquiry and topic coverage, using interaction data.
- Generated design insights for GenAI search** by prototyping and testing interfaces via Wizard-of-Oz experimentation.
- Applied **GenAI-assisted qualitative analysis** to user and AI responses to capture engagement and reasoning processes.

Revamping Higher Education in the Age of AI

Sept'24–Current

- Designed & supported user studies evaluating impact of **multimodal GenAI tutors** on students' learning outcomes, engagement and experience, focusing on **trustworthiness, safety and accuracy**; Led to **0.61 σ boost in student scores**.
- Contributed to development of campus-wide AI tutor platform at UT Austin and **generated design & engineering insights for pedagogical, data, privacy & ethical considerations** through usability testing & interviews with instructors.
- Contributed practical recommendations and design insights for rethinking, developing and evaluating higher-education assessments in the age of AI using instructor surveys, interviews and co-design sessions.

Microsoft, Redmond + University of Michigan (U-M) | UX Research Intern

Understanding Data Science Students' Needs & Misconceptions, Automatic Feedback Generation

Jan'22–July'22

- Spearheaded design and deployment** of personalized feedback generation tool for introductory data science courses at U-M, with 10-member cross-functional team of researchers, product managers, and developers across Microsoft and U-M.
- Facilitated data sharing between U-M and Microsoft**. Identified common misconceptions and core competencies on which students require support through log analysis, qualitative analysis of student data, and instructor interviews.
- Collected and analyzed user feedback**, including perceived helpfulness and impact on learning outcomes.
- Disseminated findings with executive stakeholders** at Microsoft to help improve data science developer tools.

University of Michigan | PhD Student & Graduate Researcher

Learning Experience Design, Behavioral Analysis, Aligning with Student Motivations

Jan'20–Aug'24

- Designed and analyzed **large-scale RCTs** in online data science courses and MOOCs (**N=3600+ learners**) to evaluate learning technology interventions promoting active student-AI engagement, critical thinking, and agency.
- Developed **AI-integrated personalized learning experiences** using **LLM prompting** leveraging data from past student cohorts; **Evaluated impact using metrics** such as retention, performance, student output quality and perceived value.
- Evaluated impact of interventions on student learning** through causal analysis and qualitative user feedback.
- Clustered **neural embeddings** for **interaction behavior** from ebook log-data to obtain 3 distinct behavioral groups.

IBM Research, Bengaluru | Research Software Engineer

AI for Social Good—Education & Civic Engagement

Jul'17–May'19

- Visual Assessment Generation**. Generate insights for early childhood assessment development through interviews with teachers; Leveraged **image captioning, open information extraction, and neural embedding based semantic similarity matching** to auto-generate visual assessments using knowledge base and image corpus.
- Enhancing Civic Issue Reporting**. Interviewed Indian citizens to understand usability of image, text, audio and video modalities for reporting civic issues; **Mined 1.5k images** with civic issues and associated descriptions for unsupervised training of deep learning model to **automatically detect civic issues from images**.

Selected Awards

Best Paper Awards: ACM Learning@Scale Conference, 2021 | Learning Analytics and Knowledge Conference, 2024

Awarded \$200,000 Funding for Doctoral Research: Microsoft, 2022

Summer Undergraduate Research Award: Indian Institute of Technology Delhi, 2014

Gold Medal: Indian Physics Olympiad (InPhO), Awarded to Top 35 Indian high-school students as per InPhO exam, 2012

Selected Publications [\[Google Scholar, 490+ citations, h-index: 9\]](#)

MetaCues: Enabling Critical Engagement with Generative AI for Information Seeking and Sensemaking. [A. Singh et al. arXiv preprint. 2026 \[Paper\]](#)

Enhancing Critical Thinking in Generative AI Search with Metacognitive Prompts. [A. Singh et al. Proceedings of the Association for Information Science and Technology \(ASIS&T\). 2025 \[Paper\]](#)

Protecting Human Cognition in the Age of AI. [A. Singh et al. Tools for Thought Workshop, CHI 2025 \[Paper\] \[Blog\]](#)

Impact of Multimodal and Conversational AI on Learning Outcomes and Experience. K. Taneja, [A. Singh](#), A. K. Goel. *International Conference on Artificial Intelligence in Education (AIED)*. 2026 [\[Paper\]](#)

Toward Responsible and Scalable Integration of Course-Specific AI Tutors: Instructor Experiences with a Campus-Wide Platform. G. Ko, H. K. Lee, [A. Singh](#), L. Boddy, K. Ford, and E. Huff Jr. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [\[Paper\]](#)

What's In It for the Learners? Evidence from a Randomized Field Experiment on Learnersourcing Questions in a MOOC. [A. Singh. et al. Proceedings of the Eighth ACM Conference on Learning@Scale \(L@S\). 2021. Best Paper Award \[Paper\]](#)

Bridging Learnersourcing and AI: Exploring the Dynamics of Student-AI Collaborative Feedback Generation. [A. Singh. et al. In the 14th International Learning Analytics and Knowledge Conference \(LAK\). 2024. Best Short Paper Award \[Paper\]](#)

Technical Skills & Interests

General: UX Research, Experiment Design, Data Science, Applied AI/ML/LLMs, Interaction Design, Product Development

Methods: RCTs, Field Experiments, A/B Testing, Behavioral Analysis, Log Analysis, Text Analysis, Statistical Analysis (E.g., Regression, Cluster analysis), Causal Inference, Data Visualization, Qualitative Coding, Thematic Analysis, User Studies, Usability Testing, Contextual Inquiry, Focus Groups, Surveys, Interviews, Co-design

Programming: Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-Learn, NLTK), R, SQL, OpenAI, Linux

Tools & Platforms: Jupyter, VSCode, RStudio, Github, Qualtrics, Prolific, Canva, Figma, Atlas.ti, MAXQDA

Research Areas: Cognitive Science, Learning Analytics, Human-Computer Interaction, Information Seeking & Sensemaking

Research Outreach

Invited Talk: Designing Technologies for Critical and Active Learning in the Age of AI. *EPFL*, 2025

Organised Workshop: Tools for Thought: Understanding, Protecting, and Augmenting Human Cognition with Generative AI-From Vision to Implementation. *CHI 2026 Conference*

Organised Workshop: Empowering Education with LLMs - the Next-Gen Interface and Content Generation. *Artificial Intelligence in Education Conference*, 2023

Invited Talk: Feedback Generation using Human-AI Collaboration for Introductory Data Science. *Microsoft*, 2021

Leadership & Mentoring

2020–Current: Program Committee Member/Reviewer. *Conferences:* CHI ('25,'24,'23), Learning@Scale ('25,'26), Learning Analytics and Knowledge'26, SIGCSE ('24,'23), Educational Data Mining Conference ('21); *Journals:* Computers & Education, Human-Computer Interaction, Journal of Learning Analytics

2018–Current: Mentored 5 undergraduate and 6 graduate students at the University of Michigan, UT Austin, and Georgia Tech and 3 research fellows and interns at IBM Research and Microsoft

2021: Mentored 2 teams in Educational Data Mining track, **LearnLab Summer School, Carnegie Mellon University**

2018: Organising team member for **Hackathon on AI for Social Good** at IBM Research Labs India

Teaching Experience

University of Michigan School of Information

Data Oriented Programming; Data Manipulation

Sept'20–Apr'21

Indian Institute of Technology Delhi

Probability and Stochastic Processes; Analysis and Design of Algorithms

Jul'16–May'17